

# Gauss's Law

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April 01, 2024

Let  $R$  be the integers and  $Q$  the rational numbers – we will use these letters because everything we do here will work just as well for  $R$  any UFD and  $Q$  is field of fractions (look these up). In particular, some things will seem silly, like a discussion of  $R^* :=$  the units  $:=$  the invertible elements in  $R$  (under multiplication), which of course for the integers is  $1, -1$ . But in general a UFD can have lots of units.

Def: We say non-zero  $f$  in  $R[X]$  is "primitive" if the GCD of the coefficients is 1.

**Problem 1.** Show that the product of primitives is primitive. Hint: work modulo  $p$  for each prime  $p$ .

*Proof.* Suppose  $f, g \in R[X]$  are primitives. Assume the contrary,  $f(x)g(x)$  is not a primitive. Thus the gcd of all of its coefficients is not 1 thus  $\exists p$ , prime, which divides all of its coefficients. Since,  $f$  and  $g$  are primitive,  $p$  cannot divide every coefficient of  $f$  and  $g$ . Let  $f_s x^s$  be the first term of  $f$  which is not divisible by  $p$ . Let  $g_t x^t$  be the first term of  $g$  not divisible by  $p$ . Thus  $f_s \mod p \neq 0$  and  $g_s \mod p \neq 0$ . Thus the  $r + s$ -th term of  $f(x)g(x)$  is

$$(\cdots + f_{s-2}g_{t+2} + f_{s-1}g_{t+1} + f_s g_t + f_{s+1}g_{t-1} + f_{s+2}g_{t-2} + \cdots)x^{r+s}$$

Since  $f_s$  and  $g_t$  are the first terms which are not divisible by  $p$ ,  $p \nmid f_i$  and  $p \nmid g_t$ , for  $i < s$  and  $j < t$  but  $p \nmid f_s g_t$ . Thus  $\forall i \in \{-\min s, t, \dots, \min s, t\} / 0$ ,  $p \nmid f_{s+i}g_{t-i}$ . Thus  $p$  divides each term in the sum except  $f_s g_t$ . Thus  $p$  doesn't divide the coefficient of the  $r + s$  term of  $fg$ . Thus a contradiction occurs, and  $fg$  must be primitive.  $\square$

**Problem 2a.** For each non-zero  $g$  in  $Q[X]$ , we can find  $c(g)$  in  $Q$ ,  $p(g)$  in  $R[X]$ , each unique up to multiplication by a unit, so that  $c(g)g = p(g)$  is in  $R[X]$  and primitive.

*Proof.*  $\forall g \in Q[X] \ni g \neq 0$ ,  $g = \sum_{m=0}^n \frac{a_m}{b_m} x^m \ni \gcd(a_m, b_m) = 1$  and  $b_m \neq 0$  for all  $m$ . Let  $c_g = \text{lcm}(b_1, \dots, b_n) / \gcd(a_1, \dots, a_n)$ . Thus,  $p_g = c_g g = \sum_{m=0}^n \frac{\text{lcm}(b_1, \dots, b_n)}{\gcd(a_1, \dots, a_n)} \frac{a_m}{b_m} x^m$ .

$$\begin{aligned} & \gcd\left(\frac{\text{lcm}(b_1, \dots, b_n)}{\gcd(a_1, \dots, a_n)} \frac{a_1}{b_1}, \dots, \frac{\text{lcm}(b_1, \dots, b_n)}{\gcd(a_1, \dots, a_n)} \frac{a_n}{b_n}\right) \\ &= \gcd\left(\frac{\text{lcm}(b_1, \dots, b_n)}{b_1}, \dots, \frac{\text{lcm}(b_1, \dots, b_n)}{b_n}\right) \gcd\left(\frac{a_1}{\gcd(a_1, \dots, a_n)}, \dots, \frac{a_n}{\gcd(a_1, \dots, a_n)}\right) \end{aligned}$$

$$\gcd\left(\frac{a_1}{\gcd(a_1, \dots, a_n)}, \dots, \frac{a_n}{\gcd(a_1, \dots, a_n)}\right) = \frac{\gcd(a_1, \dots, a_n)}{\gcd(a_1, \dots, a_n)} = 1.$$

Let  $p \in \mathbb{Z} \ni p \neq 1$  and  $p \neq 0$  where  $p \mid \frac{\text{lcm}(b_1, \dots, b_n)}{b_i} \forall i$ . Thus,  $b_i \mid \frac{\text{lcm}(b_1, \dots, b_n)}{p} \implies \frac{\text{lcm}(b_1, \dots, b_n)}{p} = \text{lcm}(b_1, \dots, b_n) \implies p = 1$ .

Thus,  $\gcd\left(\frac{lcm(b_1, \dots, b_n)}{b_1}, \dots, \frac{lcm(b_1, \dots, b_n)}{b_n}\right) = 1$ .

Thus,  $\gcd\left(\frac{lcm(b_1, \dots, b_n)}{\gcd(a_1, \dots, a_n)} \frac{a_1}{b_1}, \dots, \frac{lcm(b_1, \dots, b_n)}{\gcd(a_1, \dots, a_n)} \frac{a_n}{b_n}\right) = 1 \implies p_g$  is primitive.

Suppose  $\forall c_g, c'_g \in Q$  and  $\forall p_g, p'_g \in R[X]$ , such that  $c_g g = p_g$  and  $c'_g g = p'_g$  where  $p_g$  and  $p'_g$  are primitive. Thus,  $g = \frac{1}{c_g} p_g = \frac{1}{c'_g} p'_g \implies \frac{1}{c_g} p_g = \frac{1}{c'_g} p'_g$ . WLOG  $\frac{1}{c_g}, \frac{1}{c'_g} \in R$ , because we can multiply both sides by the gcd of the denominator by both sides. If  $c_g \neq 1$ , then  $\exists n \in \mathbb{Z} \ni n \mid \frac{1}{c_g} \implies n \mid \frac{1}{c'_g} p'_g$ . Since  $p'_g$  is primitive,  $n \nmid p'_g \implies n \mid \frac{1}{c'_g}$ . We divide by  $n$  from both sides, and induction reduces to the case that  $\frac{1}{c_g} = 1$  the same reasoning implies that  $\frac{1}{c'_g} = 1$  since  $\frac{1}{c'_g} \nmid p_g$  if  $\frac{1}{c'_g} \neq 1$ . Thus  $p_g = p'_g$ .  $\square$

**Problem 2b.** Show that  $c(gh) = c(g)c(h), p(gh) = p(g)p(h)$ .

*Proof.*  $c(gh)gh = p(gh)$  and  $c(g)c(h)gh = p(g)p(h)$ . Since  $p(gh)$  and  $p(g)p(h)$  are primitive, by uniqueness proven above,  $p(gh) = p(g)p(h)$ . Similarly up to uniqueness and multiplication by a unit,  $c(gh) = c(g)c(h)$ .  $\square$

**Problem 3.** Now prove "Gauss's Lemma": Let  $g, h \in R[X]$  with  $g$  primitive. Show that  $g$  divides  $h$  in  $R[X]$  iff it divides  $h$  in  $Q[X]$ .

*Proof.*  $\implies$  : Suppose  $g|h$  in  $R[X]$ . Then since  $R[X] \subset Q[X]$ ,  $g|h$  in  $Q[X]$ .

$\impliedby$  : Suppose  $g|h$  in  $Q[X]$ . Thus there exists a  $q \in Q[X] \ni gq = h$ . By the previous problem,  $\exists c_q \in Q$  and  $p(q) \in R[X]$ , where  $p(q)$  is primitive, such that  $c_q q = p(q) \implies q = \frac{1}{c_q} p(q)$ . Let  $c_q = \frac{a}{b}$  where  $a \in R$  and  $b \in R \setminus \{0\}$  and  $\gcd(a, b) = 1$ . Thus  $g(\frac{b}{a} p(q)) = h \implies bgp(q) = ha$ . Since  $\gcd(b, a) = 1, b \nmid a \implies b \mid h$ . Let  $h' = h/b$ . Thus  $gp(q) = h'a$ . Since  $g$  and  $p(q)$  are primitive,  $gp(q)$  is primitive thus  $ah'$  is primitive. But since  $h' \in R[X]$ , if  $a \neq 1$  then the gcd of all coefficients of  $ah'$  is  $a$  which is a contradiction. Thus  $a = 1$ . Thus  $gp(q) = h' \implies g|h'$  in  $R[X]$  and  $g|h'b$  in  $R[X] \implies g|h$  in  $R[X]$ .  $\square$

**Problem 4.** Prove the Eisenstein irreducibility criterion: Suppose  $f$  in  $R[X]$  is monic, and  $p$  in  $R$  is prime, suppose  $f = X^d \pmod{p}$ , and  $p^2$  does not divide  $f(0)$  (the constant term). Show  $f$  is irreducible.

*Proof.*  $\forall g \in R[X]$ , define  $\bar{g}(x) := g(x) \pmod{p}$ .

By Gauss's lemma, it suffices to show that  $f$  is irreducible in  $R[X]$ .

Assume the contrary  $f$  is reducible in  $R[X]$ . Thus  $\exists g, h \in R[X] \ni f = gh$ . Thus  $x^d = \bar{f}(x) = \bar{g}(x)\bar{h}(x)$ . This means  $\bar{g}(x) = x^n$  and  $\bar{h}(x) = x^{d-n}$ . Thus  $\bar{g}(0) = 0$  and  $\bar{h}(0) = 0$ . Thus  $p|g(0)$  and  $p|h(0)$ , thus  $p^2|g(0)h(0) \implies p^2|f(0)$ , which is a contradiction. Thus by contradiction,  $f$  must be irreducible.  $\square$

Define  $\forall g \in Q[X]$  define  $d_g := \frac{1}{c_g}$  where  $c_g g = p(g)$  and  $p(g)$  is primitive.

**Claim 5a.**  $g \in R[X]$  is irreducible in  $R[X]$  iff  $g$  is primitive and irreducible in  $Q[X]$

*Proof.*  $\implies$  : Suppose  $f \in R[X]$  is irreducible in  $R[X]$ . Assume the contrary,  $f$  is not primitive, thus there exists,  $r \in R$  where  $r \neq 1$  and  $h \in R[X]$  where  $f = rh$ . Since  $r \neq 1$ , it is not unit and thus  $f$  is not irreducible which is a contradiction. Thus  $f$  must be primitive. Assume the contrary,  $f$  is not irreducible over  $Q[X]$  thus  $\exists g, h \in Q[X]$  such that  $f(x) = g(x)h(x)$  where  $g$  and  $h$  are not in  $Q$ . Thus,

$f(x) = d_g d_h p(h) p(g) = d_{gh} p(h)(x) p(g)(x)$ . By  $f$ 's irreducibility in  $R[X]$   $d_{gh} p(g) = \pm 1$  and  $p(h) = f$  or vice versa. Argument is the same in either case. If  $d_{gh} p(g) = 1 \implies g(x) = d_g$  and  $h(x) = d_h f$ . Thus  $f$  is irreducible in  $Q[X]$ .

$\Leftarrow$  : Suppose  $f$  is primitive and irreducible in  $Q[X]$ . Then since  $R[X] \subset Q[X]$ ,  $\forall g, h \in R[X] \ni f = gh \implies g(x) = 1$  and  $h(x) = f(x)$  and vice versa since  $f$  is irreducible and primitive. Thus  $f$  is irreducible in  $R[X]$   $\square$

**Problem 5.** Prove  $R[X]$  is a UFD (look up the definition), with irreducible elements the primitive polynomials irreducible over  $Q[X]$ .

*Solution.*  $\forall f \in R[X]$ .

If  $f$  irreducible in  $Q[X]$ . By problem 2,  $\exists d_f \in R$  and  $p(f) \in R[X]$  such that  $f = d_f p(f)$ . Since  $f$  is irreducible in  $Q[X]$ ,  $p(f)$  is irreducible in  $Q[X]$  and since it is primitive, it is also irreducible in  $R[X]$ . Since  $R$  is a ufd,  $d_f$  has a unique factorisation. Thus  $f$  unique factorization is the unique factorization of  $d_f$  multiplied by  $p(f)$ . The factorization is unique because  $p(f)$  and  $d_f$  are unique.

If  $f$  is not irreducible in  $Q[X]$ . Thus there  $\exists g, h \in Q[X] \ni f = gh$ . Check if  $g$  and  $h$  are irreducible, if  $g$  or  $h$  is irreducible, use the case above for the irreducible one. If  $g$  or  $h$  is reducible, continue this case until you have gotten a irreducible polynomial and move to the case above. Reconstruct  $f$ 's unique factorisation, by multiplying the unique factorizations of  $g$  and  $h$ .  $\blacksquare$

**Problem 6.** Let  $r$  in  $\mathbb{C}$  (complex numbers) be "integral", meaning  $r$  is the root of an integer monic polynomial. Show the minimal polynomial for  $r$  over  $\mathbb{Q}$  is integer monic. Call it  $P(x)$ . Show  $R[r] = R[X]/(P(X))$ . Now take  $p$  a prime integer. Show  $R[r]/pR[r] = F_p[X]/(q(X))$  where  $q$  in  $F_p[X]$  is  $P$  modulo  $p$ . Use this to show that there is a prime ideal  $P$  in  $R[r]$  whose intersection with  $R$  is  $(p)$  (hint, recall  $p$  is maximal, so you only need one inclusion).

*Proof.* Since  $R$  are the integers now, I will just use  $\mathbb{Z}$ .

Suppose  $r \in \mathbb{C}$  is integral. Let  $f \in \mathbb{Z}[X]$  be the integer monic polynomial  $r$  is a solution to.  $\exists P \in \mathbb{Z}[X]$  a minimal polynomial for  $r$  over  $\mathbb{Q}$ , because if  $r$  is not a solution to any integer polynomial with degree less than  $f$ ,  $P = f$ . WLOG,  $p$  is primitive because if  $P$  is not primitive, use the process in 2 to get a primitive polynomial  $p(P) \in \mathbb{Z}[X]$  and since  $P$  is irreducible in  $Q[X]$ ,  $p(P)$  is also irreducible in  $Q$ . Furthermore, since  $p(P)$  is primitive and irreducible over  $Q[X]$ , it is irreducible over  $\mathbb{Z}[X]$  and thus a integer polynomial. Since  $p$  is a integer polynomial which divides a integer monic polynomial  $f$ , it must also be monic because  $\forall a, b \in \mathbb{Z}$ ,  $abx^n = x^n$  iff  $a = b = 1$ . Thus  $p$  is monic. Thus the minimal polynomial for  $r$  over  $\mathbb{Q}$  is integer monic

Since  $p$  is the minimal polynomial for  $r$  over  $\mathbb{Q}$  and is primitive, it is irreducible over  $\mathbb{Z}[X]$  and since  $r$  is a solution.  $\mathbb{Z}[r] = \mathbb{Z}[X]/(P(X))$ .  $\mathbb{Z}[r]/p\mathbb{Z}[r] = (\mathbb{Z}[X]/(P(X)))/p((\mathbb{Z}[X]/(P(X)))) = (\mathbb{Z}[X]/(p, P(X))) = (\mathbb{Z}[x]/p)/P(X) = F_p[X]/\bar{P}(X)$ ,  $\bar{P}(X) = P(x) \mod p$ , which is a finite field since  $P$  was irreducible and thus a ID meaning  $(\bar{P}(X))$  is a prime ideal.  $p\mathbb{Z} \subset p\mathbb{Z}[r]$  and since  $p$  is prime and thus maximal there is no ideal  $I$ , other than  $\mathbb{Z}$  and  $p$ , such that  $p \subset I = p\mathbb{Z}[r] \cap \mathbb{Z}[r]$ . But  $p \subseteq p\mathbb{Z}[r] \cap \mathbb{Z}[r]$ . So  $p\mathbb{Z}[r] \cap \mathbb{Z}[r] = (p)$ .  $\square$